

Secure Shell TASK – 1

- 1. Install the OpenSSH server package on your Linux system and verify that the sshd service is running by connecting to localhost using the ssh command.**

Ans :

- `sudo dnf install openssh-server -y`
- `sudo systemctl enable --now sshd`
- `sudo systemctl status sshd`
- `ssh your_username@localhost`

- 2. Generate a new SSH key pair using ssh-keygen, and add your public key to the ~/.ssh/authorized_keys file of your user account to enable key-based authentication.**

Ans :

- `# Generate a new SSH key pair`
- `ssh-keygen -t rsa -b 4096`

- `# Create the .ssh directory if it doesn't exist`
- `mkdir -p ~/.ssh`

- `# Add the public key to the authorized_keys file`
- `cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys`

- `# Set the correct permissions`
- `chmod 700 ~/.ssh`
- `chmod 600 ~/.ssh/authorized_keys`

- `# Test key-based SSH authentication`
- `ssh localhost`

3. Edit the /etc/ssh/sshd_config file to disable password authentication by setting PasswordAuthentication no, then restart the sshd service and test that you can only log in using your SSH key.

Ans :

- # Edit the SSH server configuration file
- sudo nano /etc/ssh/sshd_config
- # Find and change the following line:
- PasswordAuthentication no
- # Save and exit (Ctrl + O, Enter, Ctrl + X)
- # Restart the SSH service
- sudo systemctl restart sshd
- # Check the service status
- sudo systemctl status ssh
- # Test SSH login using your key
- ssh localhost

**4. Document each step you took to configure SSH key-based authentication and disable password login, including the exact commands and configuration changes you made.

Hint: Use screenshots or command outputs to support your documentation.**

Ans :

Step 1: Install OpenSSH Server

Command:

sudo apt update

```
sudo apt install openssh-server -y
```

Verification:

```
sudo systemctl status ssh
```

Expected Output:

Active: active (running)

Step 2: Generate an SSH Key Pair**Command:**

```
ssh-keygen -t rsa -b 4096
```

Press **Enter** to accept the default file location and optionally enter a passphrase.

Step 3: Configure Key-Based Authentication**Commands:**

```
mkdir -p ~/.ssh
```

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

```
chmod 700 ~/.ssh
```

```
chmod 600 ~/.ssh/authorized_keys
```

These commands create the SSH directory (if necessary), add the public key to the `authorized_keys` file, and set the correct permissions.

Step 4: Test SSH Key Authentication**Command:**

ssh localhost

Step 5: Disable Password Authentication

Edit the SSH server configuration file.

Command:

```
sudo nano /etc/ssh/sshd_config
```

Locate the following line and change it to:

```
PasswordAuthentication no
```

Save the file and exit.

Configuration Change:

```
PasswordAuthentication no
```

Step 6: Restart the SSH Service

Command :

```
sudo systemctl restart ssh
```

```
sudo systemctl status ssh
```

Expected Output:

```
Active: active (running)
```

Step 7: Verify Password Login is Disabled

Command:

```
ssh -o PubkeyAuthentication=no localhost
```

Expected Output:

```
Permission denied (publickey).
```
